

Data sharing in PredRet for accurate prediction of retention time: Application to plant food bioactive compounds

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ABSTRACT

Prediction of retention times (RTs) is increasingly considered in untargeted metabolomics to complement MS/MS matching for annotation of unidentified peaks. We tested the performance of PredRet (<http://predret.org/>) to

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predict RTs for plant food bioactive metabolites in a data sharing initiative containing entry sets of 29–103 compounds (totalling 467 compounds, >30 families) across 24 chromatographic systems (CSs). Between 27 and 667 predictions were obtained with a median prediction error of 0.03–0.76 min and interval width of 0.33–8.78 min. An external validation test of eight CSs showed high prediction accuracy. RT prediction was dependent on shape and type of LC gradient, and number of commonly measured compounds. Our study highlights PredRet's accuracy and ability to transpose RT data acquired from one CS to another CS. We recommend extensive RT data sharing in PredRet by the community interested in plant food bioactive metabolites to achieve a powerful community-driven open-access tool for metabolomics annotation.

1. Introduction

The dark matter in metabolomics refers to the large fraction of molecular signals that are detected with untargeted analyses but remain unidentified. Part of this dark matter corresponds to the food metabolome. Currently, >26,000 compounds have been described in foods (<https://foodb.ca>), and upon ingestion and digestion, these food components are further transformed into various metabolites (Scalbert et al., 2014), many of which are not identified or inventoried yet in databases (Barabási, Menichetti, & Loscalzo, 2020). Plant food bioactive compounds (also referred as dietary phytochemicals, e.g., (poly)phenols, carotenoids, glucosinolates, alkaloids) and their phase I, -II and gut microbial metabolites represent an important class of the food metabolome that receive widespread interest for their protective health effects and more recently, for their usefulness as food intake biomarkers. They cover a large chemical space ranging from highly polar to lipophilic compounds, and their identification in untargeted methods remains a challenging feat.

Identification of unknowns in untargeted metabolomics combines multiple types of information and tools, such as matching of exact mass in compound databases, comparison of experimental to reference MSⁿ spectral data and chromatographic retention time (RT) of authentic standards for Metabolomics Standards Initiative (MSI) level I identification, or to publicly available spectral databases for MSI level II (Sumner et al., 2007). But searches in databases often return an excessive number of structurally similar hypotheses (Hall et al., 2018), and purchasing all corresponding standards is not feasible due to limited availability and high cost. In the case of plant food bioactive compounds and their metabolites, identification is further challenged by the lack of commercial standards and the high structural similarity between many isomeric compounds, which makes their MS/MS spectra indistinguishable.

Leveraging orthogonal data such as RT becomes valuable for assisting the certainty of identification to MSI levels I and II, by narrowing the number of plausible hypotheses within an observed RT window. Recent years have seen several approaches to adopt RT prediction models for integration into untargeted analysis workflows with varying degrees of success (McEachran et al., 2018; Witting & Böcker, 2020). Existing types of RT prediction models include i) simple algorithms based on log P or gradient back-calculation (Boswell, Schellenberg, Carr, Cohen, & Hegeman, 2011; Abate-Pella et al., 2015), ii) monotonically constrained generalised additive model (GAM) (Stanstrup, Neumann, & Vrhovšek, 2015) of retention times and iii) complex *in silico* quantitative structure-retention relationship (QSSR) models based on combinations of molecular descriptors. QSSRs can be built using different machine learning approaches, such as artificial neural network, random forest and support vector regression models (Aalizadeh, Nika, & Thomaidis, 2019; Domingo-Almenara et al., 2019; Hall et al., 2018; McEachran et al., 2018; Bouwmeester, Martens, & Degroove, 2020; Naylor, Catrow, Maschek, & Cox, 2019; Bade et al., 2015; Tada et al., 2019; Wolfer et al., 2015). However, these prediction models are limited in their application, as RT data are specific to one chromatographic system (CS) and the models do not provide accurate predictions outside the trained conditions.

As analytical methods are not harmonised and most laboratories tend

to have their own routine semi-targeted or untargeted LC methods for covering plant food bioactive compounds in various types of matrices (serum, plasma, urine, digestive fluids, food materials), it is ideal that RT prediction models be customisable across CSs. PredRet (Stanstrup et al., 2015) represents an original approach that enables users of the scientific community to benefit from RT data sharing through its open access RT database, and obtain predictions in their own CS if the RT of a compound has been experimentally determined by another user or laboratory. In this aspect, PredRet is relevantly applicable for transposing RTs between CSs differing in mobile phase composition, gradient, flow rate and column dimensions. In the framework of the COST Action POSITIVE (<https://www6.inra.fr/cost-positive>, FA1403), we evaluated the performance of PredRet to predict the RTs of plant food bioactive compounds and their metabolites in a multi-laboratory test involving 19 laboratories across Europe, using 24 gradient-based reversed-phase CSs. We also expanded PredRet database with experimental RTs of 467 plant food compounds.

2. Experimental section

2.1. Chemical compounds

All participating laboratories purchased their own chemicals, differing from one laboratory to another, except that 10 laboratories previously involved in a multiplatform coverage test organised by the COST Action POSITIVE, received two common standard mixtures comprising of 56 plant food bioactive compounds (Koistinen et al., 2018). Synthesised standards ($n = 49$) were accepted in addition to commercial standards, provided that the structure was unambiguously elucidated by NMR and MS/MS spectra and that the compounds are entered in the online platform for food compound exchange, FoodComEx (<https://foodcomex.org/>). Depending on laboratories, chemicals were analysed in solvent or spiked in biological matrices (urine or plasma). A full list of the 467 analysed compounds is provided in Table S1, with their common name, InChI, IDs in HMDB, FoodDB and PhytoHub, taxonomy, chemical structure, formula, monoisotopic mass, predicted logP and the number of CSs where they were analysed.

Experimental RT datasets containing compound name, InChI and/or chemical structure were provided by the involved laboratories. InChIs were used as unambiguous identifiers for recognition of identical compounds between CSs and compound names were harmonised across laboratories. For polyphenol metabolites, we applied the new reference KCC nomenclature (Kay et al., 2020). InChIs were either extracted from databases such as PhytoHub (<http://phytohub.eu>), PubChem (<https://pubchem.ncbi.nlm.nih.gov>) (Kim et al., 2019), HMDB v4.0 (www.hmdb.ca) (Wishart et al., 2018) or computed from chemical structures using Marvin v19.7, 2019, ChemAxon (<https://www.chemaxon.com>). LogP values were computed using ALOGPS v2.1 (<http://www.vcclab.org/lab/alogps/>) (Tetko et al., 2005; VCCLAB, 2005) after conversion of InChIs to SMILES via InChIToSMILES (<http://www.chemspider.com/inchi.asm.x>) (Pence & Williams, 2010). In PredRet database, the main InChI layer containing chemical formula, atom connections and hydrogen atom sublayers is considered when matching compounds, and information after the main layer (e.g., charge, stereochemical and isotopic layers) is ignored.

2.2. Chromatographic systems

Experimental RT data were collected from 24 CSs across 19 laboratories. These CSs were not intentionally optimised for the RT prediction test but rather represent the routine semi-targeted or untargeted metabolomic methods of the various laboratories. A full description of instrument, column and analytical conditions used in the 24 CSs is provided in Table 1. Overall, 15 C18 reverse-phase (RP) columns from various manufacturers were used with dimensions ranging from 0.5 to 4.6 mm (internal diameter), 50 to 250 mm (length) and 1.6 to 5 μ m (particle size). HPLC or UHPLC methods were used in acidic conditions. Water and acetonitrile acidified with formic acid (0.1–0.9%) or trifluoroacetic acid (0.1%) were most commonly used as mobile phases A and B, while three CSs used methanol or acetone as mobile phase B. The gradients utilised in 13 UHPLC methods consisted of linear and multi-phasic slopes with flow rates of 0.4 to 0.6 mL/min and total run times ranging from 6 to 26 min. There were four HPLC methods with multi-phasic slopes with flow rates of 0.015 to 1.5 mL/min and longer run times of 20 to 135 min. Fig. S1 shows the diversity of gradient slopes in the 24 CS.

2.3. Prediction of retention times

Experimentally measured RTs (Table S3) were entered in PredRet for the 467 compounds listed in Table S1. The number of measured RTs by CS varied from 29 in CS9 to 103 in CS14. For each CS, the compound names, InChI and experimentally measured RTs were entered into PredRet web interface (<http://predret.org>) along with a description of the respective CS method. PredRet is then able to predict RTs for compounds that have not been previously experimentally measured in one CS but have been determined in some other CS. The prediction is achieved by constructing GAMs between all pairs of CSs in the PredRet database using the compounds that were measured in both CSs. Empirical prediction intervals (PI) were established via bootstrapping of GAMs, as described in more details by Stanstrup et al (2015). The model providing the prediction with narrowest PI was then used. Predictions were flagged as suspicious by the program if the RT is considered potentially incorrect, when the difference between experimental and predicted RTs was $\geq$ twice the distance from the predicted RT to outer limits of the PI. Predictions were automatically discarded if their PI widths were $\geq$ 2 min or $\geq$ 20% of the predicted RT. The total number of RT predictions between CSs, as well as accuracy and coverage of PI relative to the total chromatographic run time, were compared.

2.4. Validation of predicted retention times

A validation test was conducted on CSs 1, 2, 4, 5, 14, 18, 19, and 22, which had the highest number of experimental RT values. These eight CSs comprise of UHPLC and UPLC methods varying in LC instrument and gradient, column, mobile phases, flow rate and run time. The experimental RT datasets of these CSs were split into training sets (80% data, $n = 79, 71, 73, 67, 82, 63, 63$, and 78 compounds respectively) and test sets (20% data, $n = 20, 18, 18, 17, 21, 16, 16$, and 19 compounds respectively). For selection of compounds in the test sets, the datasets were split into three equal sections covering the beginning, middle and end of the chromatographic run, and then 20% of the compounds were randomly selected from the three sections to ensure a uniform distribution of RT along the entire chromatographic run. Another criterion was to select, in the test set, the same proportion of unique compounds as in the whole dataset of the selected CSs. Validation of RT predictions for each of the eight selected CSs was performed in conditions where the complete datasets of the remaining 23 CSs were entered into the PredRet database.

3. Results and discussion

3.1. Large diversity of plant food metabolites analysed

A total of 1583 experimental RT values were collected for 467 plant food compounds or related human metabolites in one or several of the 24 CSs used by the 19 participating platforms. The 467 compounds belong to > 30 families including flavonoids (anthocyanins, flavonols, flavones, flavanols, flavanones, isoflavones), phenolic acids, lignans, ellagitannins, coumarins and furanocoumarins, nitrogen-containing compounds (i.e., alkaloids, amines, indoles), glucosinolates, alkylresorcinols, thiosulfates, tocopherols, phytosterols, carotenoids and mono, di-, sesqui- and triterpenoids, and their human metabolites, e.g., glucuronidated and sulfated conjugates, as well as gut microbial metabolites. They cover a large chemical space from highly polar to lipophilic with predicted logP values from -3.48 to 10.40 and with monoisotopic masses from 95.0371 to 934.0712 Da (Fig. 1). The PredRet database is growing continuously with addition of new compounds and associated RT data by registered users. At the time of our experiment, a limited number of plant food compounds was present in PredRet, and our datasets represented a major update for this category of compounds.

The number of CSs in which each compound was analysed is provided in Table S1. Of the 467 entered compounds, 212 were analysed in one CS only, while 4'-hydroxy-3'-methoxycinnamic (ferulic), 4-hydroxy-3-methoxybenzoic (vanillic), 3,4-dihydroxybenzoic (protocatechuic), 5-O-caffeoylquinic and 4'-hydroxycinnamic (*p*-coumaric) acids were most commonly measured in 20 of the 24 CSs (Fig. S2). The size of the datasets varied from 29 to 103 experimental RTs. CSs 1, 2, 4, 5, 7, 14, 17, 18, 19, 22, and 23 contained ≥ 75 RTs, as illustrated by their large node size in Fig. 2, in contrast to CS9 and CS16, which contained the least RT data (29 and 35 RTs, respectively). Across the platforms, CSs 2, 6, 11, 13, and 15 shared the highest compound overlap as evidenced by their highly connected nodes (Fig. 2) while still showing relatively good overlap with CSs 1, 3, 7, 14, 16, 22, and 23. Pairwise clusters of CSs 18–19 and 4–5 were observed as they shared $>90\%$ compounds similarity, corresponding to two analytical methods from the same platform.

3.2. Retention time prediction coverage and rate

A total of 6382 new RT predictions were obtained for the 24 CSs, with up to 667 predictions for one CS (Table 2 and Fig. S3). Compounds that were entered in PredRet prior to this study (1783 unique compounds, $\sim 10\%$ were plant food bioactive compounds) contributed to prediction of additional compounds beyond the 467 compounds entered in this study. We observed a general trend that as more experimental RTs are entered in PredRet, more RT predictions are generated for compounds not previously analysed. This is demonstrated in CSs 1, 2, 22, and 23 where 559, 539, 667, and 572 new RT predictions were generated from 98, 89, 97 and 75 compounds entered into PredRet respectively (Table 2). However, RT prediction was also dependent on shape (Fig. S1) and type (i.e., UHPLC or HPLC) of the LC gradient as well as number of common compounds shared with other CSs. For example, infrequently used mobile phases may limit the predictability of a CS. The entry of 29 compounds for CS9 was not sufficient to obtain RT predictions. However, despite relatively small RT datasets (35 to 46 compounds) were entered for CSs 11, 15 and 16, they had a high prediction rate, explained by a versatile CS and/or good combination of compounds.

3.3. Retention time prediction accuracy

PredRet provided RT predictions for compounds never analysed in the CSs but also for compounds in the entry dataset. We used the latter to compare prediction accuracy between CSs. RT predictions were highly accurate across the 24 CSs, with median prediction errors between 0.03 and 0.76 min (Table 2). As run times vary greatly across CSs (5 to 135

Table 1
Instrument and conditions of chromatographic systems used by participating platforms.

Method	LC instrument	Column specifications	Column temperature (°C)	Mobile phases	sample matrix	Flow rate (mL/min)	Run time (min)	LC gradient (t[<i>min</i>], %B)
U-CS1	UHPLC: Thermo U3000 QTOF MS: Bruker Impact HD2	Waters Acquity HSS T3 (2.1x150 mm, 1.8 µm, 100 Å)	30	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	solvent, plasma	0.4	26	(0, 0), (2, 0), (15, 100), (22, 100), (22.1, 0), (26, 0)
U-CS2	UHPLC: Agilent 1290 QTOF MS: Agilent 6540	Agilent Zorbax Eclipse XDB C18 (2.1x100 mm, 1.8 µm, 80 Å)	50	A: H ₂ O + 0.1% FA B: MeOH + 0.1% FA	plasma	0.4	16.5	(0, 2), (10, 100), (14.5, 100), (14.51, 2), (16.5, 2)
U-CS3	UHPLC: Thermo U3000 QTOF MS: Bruker Impact HD2	Waters Acquity UPLC BEH Shield RP18 (2.1x100 mm, 1.7 µm, 130 Å)	30	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	solvent, urine	0.6	26	(0, 0), (2, 0), (7, 10), (22, 95), (22.1, 0), (26, 0)
U-CS4	UHPLC: Agilent 1290 Infinity QTOF MS: Agilent 6550 iFunnel	Agilent Zorbax Eclipse Plus RRHD (2.1x50 mm, 1.8 µm, 95 Å)	30	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	plasma, urine	0.4	12	(0, 1), (5, 10), (8, 25), (9.1, 99), (10, 99), (12, 1)
U-CS5	UHPLC: Thermo Accela 1250 QTRAP MS: Thermo Exactive	Agilent Zorbax Eclipse Plus RRHD (2.1x50 mm, 1.8 µm, 95 Å)	30	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	plasma, urine	0.4	12	(0, 1), (5, 10), (8, 25), (9.1, 99), (10, 99), (12, 1)
H-CS6	HPLC: Agilent 1260 QTOF MS: Agilent 6530	Phenomenex Synergi Hydro-RP (2x150 mm, 4 µm, 80 Å)	30	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	solvent, urine	0.5	45	(0, 5), (1, 5), (35, 45), (40, 100), (45, 100)
U-CS7	UHPLC: H-class QTOF MS: Synapt G2 S	Waters Acquity UPLC BEH Shield RP18 (2.1x150 mm, 1.7 µm, 130 Å)	40	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	solvent, urine	0.4	37.1	(0, 5), (30, 50), (31, 100), (37, 100), (37.1, 0)
H-CS8	HPLC: Waters HPLC 2695 PDA: Waters 2996	Interchim Supelcosil LC-18 (4.6x250 mm, 5 µm, 120 Å)	40	A: H ₂ O + 0.1% TFA B: ACN + 0.1% TFA	solvent	1	50	(0,5), (45, 35), (47, 75), (49, 35), (50, 5)
H-CS9	HPLC: Agilent 1200 QTRAP MS: AB Sciex 4000	Phenomenex Kinetex PFP (4.6x100 mm, 2.6 µm, 100 Å)	35	A: H ₂ O + 0.1% TFA B: ACN + 0.1% TFA	solvent	1.5	32	(0, 1), (7, 7.5), (14, 7.6), (17, 10), (18.5, 12), (20, 12.5), (24, 30), (25, 90), (25.1, 1), (32, 1)
H-CS10	HPLC: Waters Alliance 2695 QTOF MS: Waters Premier	Waters Atlantis T3 (2.1x100 mm, 3 µm, 100 Å)	40	A: H ₂ O + 0.1% TFA B: ACN + 0.1% TFA	solvent	0.3	25	(0, 10), (1, 10), (6, 40), (7, 50), (8, 50), (14, 70), (16, 80), (18, 80), (20, 10), (25, 10)
U-CS11	UHPLC: Agilent 1290 QTRAP MS: Sciex 6500	Phenomenex Luna Omega Polar C18 (2.1x100 mm, 1.6 µm, 100 Å)	40	A: H ₂ O + 0.5% FA B: ACN + 0.5% FA	solvent, urine	0.5	7	(0, 5), (3, 50), (3.1, 100), (5, 100), (5.1, 5), (7, 5)
U-CS12	UHPLC: Agilent 1290 QTRAP MS: Sciex 6500	Phenomenex Luna Omega Polar C18 (2.1x100 mm, 1.6 µm, 100 Å)	40	A: H ₂ O + 0.1% FA + 10 mM NH ₄ COOH, B: ACN	solvent, urine	0.5	14	(0, 5), (8, 20), (10, 100), (12, 100), (12.1, 5), (14, 5)
H-CS13	HPLC: Agilent 1200 QTOF MS: Agilent G6530A	Phenomenex Luna C18 (4.6x150 mm, 3 µm, 100 Å)	25	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	solvent, urine	0.5	65	(0, 0), (30, 30), (35, 40), (50, 80), (52, 80), (60, 0), (65, 0)
H-CS14	HPLC: Agilent 1290 QTOF MS: Agilent 6550	Agilent Poroshell 120 EC C18 (3x100 mm, 2.7 µm, 120 Å)	25	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	solvent, urine	0.4	30	(0, 5), (10, 25), (20, 40), (24, 90), (25, 90), (26, 5), (30, 5)
H-CS15	HPLC: Eksigent nanoLC QTOF MS: Sciex TripleTOF 6600	Eksigent HALO C18 (0.5x50 mm, 2.7 µm, 90 Å)	35	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	solvent, urine	0.01	16	(0, 5), (12, 95), (14, 95), (16, 5)
H-CS16	HPLC: AB Sciex MicroLC 200 QTOF MS: AB Sciex 6500+	Eksigent HALO C18 (0.5x100 mm, 2.7 µm, 100 Å)	45	A: H ₂ O + 0.9% FA B: ACN + 0.9% FA	solvent, urine	0.015	5	(0, 1), (0.5, 1), (4, 95), (4.5, 1), (5, 1)
H-CS17	HPLC: Agilent 1290 QTOF MS: Agilent 6520	Phenomenex Synergi Hydro (2x250 mm, 4 µm, 80 Å)	25	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	solvent	0.2	50.1	(0, 0.5), (7, 0.5), (12, 12.5), (25, 16.3), (47, 35), (48, 65), (50, 65), (50.1, 0.5)
U-CS18	HPLC: Dionex Ultimate 3000 FT Orbitrap LTQ-XL MS: Thermo	Phenomenex Kinetex Core shell (2.1x150 mm, 2.6 µm, 100 Å)	40	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	plasma	0.35	15	(0, 0), (1, 0), (12.5, 100), (14, 100), (14.2, 0), (15, 0)

(continued on next page)

Table 1 (continued)

Method	LC instrument	Column specifications	Column temperature (°C)	Mobile phases	sample matrix	Flow rate (mL/min)	Run time (min)	LC gradient (t[<i>min</i>], %B)
U-CS19	HPLC: Dionex Ultimate 3000 FT Orbitrap LTQ-XL MS: Thermo	Phenomenex Kinetex Core shell (2.1x150 mm, 2.6 μ m, 100 Å)	40	A: H ₂ O + 0.1% FA B: ACN + 0.1% FA	urine	0.35	12	(0, 0), (1, 5), (7, 45), (8.5, 80), (10.5, 80), (11, 5), (12, 5)
H-CS20	HPLC: Shimadzu Prominence System PDA: SPD-M20A	Phenomenex Kinetex PFP (4.6x100 mm, 5 μ m, 100 Å)	45	A: H ₂ O + 0.1% TFA B: MeOH	solvent	0.6	20	(0, 40), (20, 72), (21, 40)
H-CS21	HPLC: Waters Alliance 2695 QqQ-MS: Micromass® Quattro Micro	LiChrospher®100 LiChroCART® (4x250 mm, 5 μ m, 100 Å)	35	A: H ₂ O + 0.5% FA B: ACN + 0.5% FA	solvent	0.3	135	(0, 5), (10, 5), (30, 15), (45, 20), (65, 20), (95, 54), (110, 63), (115, 5), (135, 5)
U-CS22	UHPLC: Waters Acquity QTOF MS: Waters Premier	Waters Acquity BEH C18 (2.1x100 mm, 1.7 μ m, 130 Å)	65	A: H ₂ O + 0.1% FA B: 80% ACN + 20% Ac + 0.1% FA	solvent, plasma	0.4	6	(0, 0), (5, 100), (5.5, 0), (6, 0)
U-CS23	UHPLC: Waters Acquity QTOF MS: Waters Premier	Waters Acquity HSS T3 C18 (2.1x100 mm, 2.6 μ m, 100 Å)	50	A: H ₂ O + 0.1% FA B: 70% ACN + 30% MeOH + 0.1% FA	solvent, plasma, urine	0.5–1.2	7	(0, 5), (1, 8), (2, 15), (3, 40), (4, 70), (4.5, 100), (5, 100), (6.4, 100), (6.6, 5), (6.8, 5), (7, 5)
U-CS24	UHPLC: Dionex Ultimate 3000 QqQ-MS: Thermo Fisher TSQ Vantage	Phenomenex Kinetex EVO C18 (2.1x100 mm, 2.6 μ m, 100 Å)	40	A: H ₂ O + 0.2% FA B: ACN + 0.2% FA	plasma, urine	0.4	12	(0, 5), (0.5, 5), (7, 95), (8, 95), (8.5, 5), (12, 5)

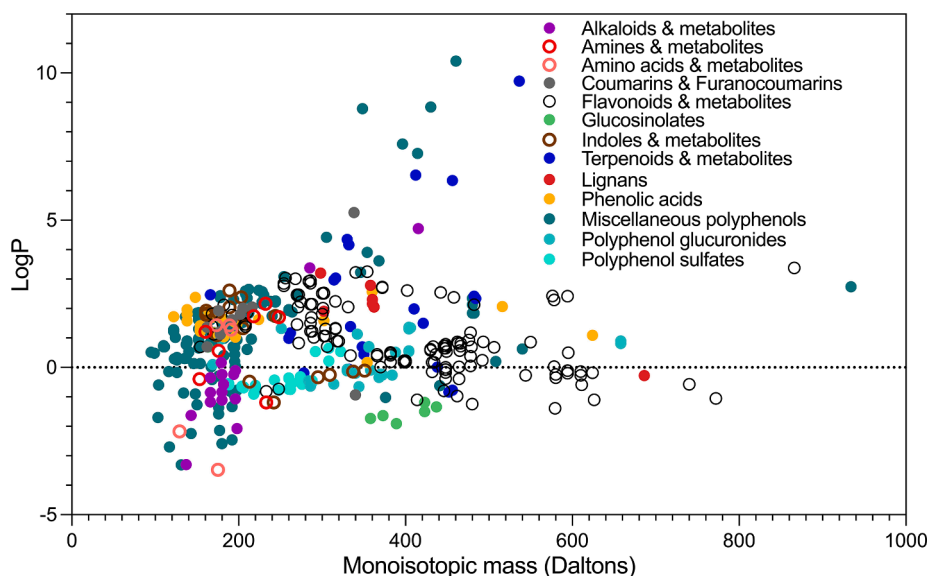


Fig. 1. Chemical space covered by the 467 plant food metabolites entered in PredRet.

min), median prediction errors were also expressed in percentage relative to the total runtime, ranging from 0.3% to 1.8% (CS9 excluded).

A graph comparing experimental and predicted RTs for compounds of CS1 entry dataset is given in Fig. 3 as an example. Equivalent graphs for all other CSs are provided in Fig. S4. In CS1, accurate predictions with narrow PI were obtained for most compounds with RT ranging between 6.6 and 14.2 min. Predictions for eight compounds (myo-inositol, proline betaine, dopamine, 3,4,5-trihydroxybenzoic acid (gallic acid), 1,3-dimethyluric acid, α -tocopherol, ursolic acid and alkylresorcinol C17:0) were discarded by PredRet algorithm as their PI widths were ≥ 2 min or $>20\%$ of the predicted RT. PI width is an important indicator of prediction accuracy as it represents how accurate the projection models are, based on the number of experimentally known RTs in the RT range of compounds that are being projected in the pairwise CS

models (Stanstrup et al., 2015). We observed that predictions were usually missing at the beginning and end of the runs, where there tends to be a low density of known RTs, and conditions are approaching the analytical limits of the CSs (Fig. S5). In CS1, predictions were not generated before the first 1.5 min and after 14 min. For 15 compounds (1-methylpiperidine, arbutin, 1-methylxanthine, 1*H*-pyrrole-2-carboxaldehyde, cyclo(Leu-Pro), 5-(3',4'-dihydroxyphenyl)valeric acid, homoeriodictyol, tomatidine, formononetin, bergapten, nobiletin, isosakuranetin, kaempferide, biochanin A, and bergamottin), RT prediction was not expected, as they were not present in any other CS. Globally, PredRet performed well for CS1 with a median prediction error of 0.07 min (0.27% of runtime) and median PI width of 0.83 min. For 77 non-unique compounds entered into PredRet, 559 new predictions for compounds never analysed in this system were obtained, in the range of

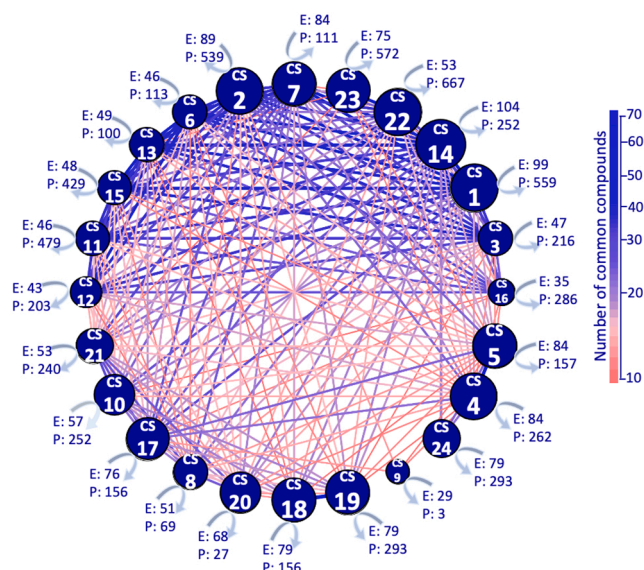


Fig. 2. Network map illustrating compound coverage overlap. Size of node represents the number of compounds present in the dataset while the thickness and colour of edges represent the number of common compounds between the paired datasets. The thicker the edge, the larger number of common compounds. Edge colours — and — denote low (<10) and high (>60) similarity of compounds, respectively. E: number of RT data entered into PredRet; P: number of new RT predictions made.

1.01 to 14.18 min.

Amongst CS7, CS8, CS9, CS14, CS17, CS20, and CS24, a common trait is the high proportion of rare plant compounds unique to their CSs, which indirectly resulted in a low number of common compounds shared with other CSs. For example, CS7 contained sesquiterpenoids not represented in other CSs, likewise for anthocyanin glycosides in CS8, urolithins and conjugated isoflavone metabolites in CS14, glucosinolates and rare flavonoids in CS17, flavonolignans (e.g., dehydrosilydianin), rare flavonoids and sulfated conjugates in CS20, and urolithins and conjugated flavonoids in CS24. Adding the RTs of these rare plant food compounds and metabolites contributes to the richness of PredRet database; however, a caveat is that these CSs themselves may not receive the benefit of good prediction coverage. In such circumstances, the user is encouraged to include common plant compounds that are also frequently represented in the PredRet database. As an example, we propose a list of 14 compounds frequently analysed in our study ($\geq 67\%$ of 24 CSs), which covers a wide RT range: 4'-hydroxy-3'-methoxycinnamic acid (ferulic acid), 4-hydroxy-3-methoxybenzoic acid (vanillic acid), 5-O-caffeoylquinic acid, 4'-hydroxycinnamic acid (*p*-coumaric acid), 3,4-dihydroxybenzoic acid (protocatechuic acid), 3',4'-dihydroxycinnamic acid (caffeic acid), 3,4,5-trihydroxybenzoic acid (gallic acid), 3',5'-dimethoxy-4'-hydroxycinnamic acid (sinapic acid), (–)-epicatechin, kaempferol, hippuric acid, luteolin, phloretin, and hesperetin (Table S4).

To further validate the predictive performance of GAM in PredRet, we performed an external validation test on a subset of eight CSs, splitting the experimental datasets into 80% for training sets and 20% for test sets. The training sets were used to build GAMs between CSs in PredRet database to obtain predictions with PIs for the compounds in the test sets. Predictions were compared to experimental data to obtain the prediction error for each compound (Table S5) and the prediction statistics for each CS are provided in Table S6. Accurate predictions were achieved, with the median prediction error in the test sets ranging from 0.04 to 0.41 min across the eight CSs. The maximum absolute prediction error was 3.55 min for α -tocopherol (CS2), followed by catechol (2.45 min, CS5). It is difficult to compare the performance of PredRet with other RT prediction tools as those only allow predictions within the

same CS, while PredRet predicts RTs from one CS to another CS differing in mobile phase composition, gradient and flow rate.

Despite accurate models being built for the CSs, we observed that early- and late-eluting compounds were generally omitted from predictions, likely due to their extreme polarity. Compounds unique to respective CSs (e.g., nobiletin in CS1, 2',5'-dihydroxyphenylacetic acid in CS2 and 9-hydroxy-urolithin-3-glucuronide (isourolithin A glucuronide) in CS14) did not obtain RT predictions as well as compounds that did not have sufficient RT data density in the RT area (e.g., *N*-(3-hydroxybenzoyl)glycine (2-furoylglycine) in CS19 and pinoresinol in CS22). Between 14 and 39% of the compounds in the CSs of the validation (test) set had experimental RTs that fall outside the estimated PI, showing that the PIs should be interpreted with caution, as previously noted in the original paper describing PredRet. The practical implication is that a proposed annotation for an experimental RT cannot be completely discarded even if the RT falls outside the proposed annotation's PI. A few limitations of PredRet were identified in our study that may be corrected in the future. Firstly, users have no information about the standards that have been considered for providing predictions in their CS: e.g., commercial or synthesised standard, analysed in solvent or spiked in a biological matrix. Secondly, PredRet algorithm recognises the entered compounds based on the main InChI layer only and therefore stereochemical information is ignored during RT prediction.

3.4. Application of PredRet predictions for identification of plant food compounds in metabolomic studies

The effectiveness of RT prediction using PredRet allowed the distinction of isomeric compounds. In Fig. 4A, 3-(3',4'-dihydroxyphenyl)propanoic (dihydrocaffeic) acid with a predicted RT of 8.3 min (PI: 8.1 to 8.5 min) could be distinguished from its isomers, 4'-hydroxy-3'-methoxyphenylacetic acid (homovanillic) acid (PI: 8.5 to 9.0 min) and 3,4-dimethoxybenzoic acid (veratric) acid (PI: 9.3 to 9.6 min). In Fig. 4B, the predicted RTs of fisetin (PI: 9.8 to 10.7 min), kaempferol (PI: 11.4 to 12 min) and luteolin (PI: 10.6 to 11.3 min) were also clearly distinguished, except for the narrow overlap in the PIs of luteolin and fisetin (10.6 to 10.7 min). This is particularly useful as an orthogonal parameter to eliminate hypotheses when identifying unknown features with the same *m/z* in untargeted metabolomics studies. In addition, as RTs of flavonoid conjugates (glycosides, glucuronides) differ from that of their aglycones, prediction of RT may help to distinguish between aglycones truly present in the samples and detected aglycones that are generated during the analysis as in-source fragments of glycosides or glucuronides.

Another useful application of PredRet is aiding in annotation of rare plant food compounds in untargeted metabolomics studies, when the standards are not commercially available or difficult to synthesise. As soon as a user enters experimental data for a rare plant food compound in a CSs, PredRet provides RT prediction with PI for this compound in CSs where it has not been experimentally measured. For example, the contribution of tomatidine's experimental RT (11.8 min) from CS1 enabled the prediction of RTs in 15 other CSs, while formononetin (CS1), 8-hydroxy-urolithin-3-sulfate (CS14) and 8-deoxylactucin (CS7) enabled the prediction of RTs in 13 other CSs. To optimise this process, it is crucial that users who entered experimental RTs for rare compounds also enter experimental RTs for common compounds such as those suggested above.

4. Conclusion

PredRet, based on pairwise GAMs, was demonstrated to be a useful tool for obtaining a good number and highly accurate RT predictions for plant food bioactive compounds and their metabolites. Its use in untargeted metabolomics studies can definitely help for tentative identification, by eliminating hypotheses that do not fall within the predicted RT range, or when commercial standards are not readily

Table 2
Statistics of PredRet retention time predictions for 24 liquid chromatographic systems (CSs) with an entry dataset of 467 plant compounds.

Prediction statistics	CS 1	CS 2	CS 3	CS 4	CS 5	CS 6	CS 7	CS 8	CS 9	CS 10	CS 11	CS 12	CS 13	CS 14	CS 15	CS 16	CS 17	CS 18	CS 19	CS 20	CS 21	CS 22	CS 23	CS 24
# Experimental RTs (# non-unique compounds)	98 (76)	89 (78)	47 (47)	90 (88)	83 (83)	46 (46)	84 (65)	51 (44)	29 (22)	57 (47)	46 (45)	43 (43)	49 (49)	103 (75)	48 (48)	35 (35)	75 (50)	79 (79)	79 (79)	68 (38)	53 (46)	97 (83)	75 (59)	54 (40)
# Predictions made # Predictions made where experimental RTs are unknown	623 559	603 539	245 216	323 262	215 157	152 113	157 111	91 69	4 3	309 272	518 479	241 203	131 100	309 252	466 429	314 286	198 156	387 313	366 293	45 27	278 240	744 667	627 572	96 67
Total runtime (min)	26	16.5	26	12	12	45	37.1	50	32	25	7	14	65	30	16	5	50.1	15	12	20	135	6	7	12
Median prediction error ¹ (min)	0.07	0.10	0.19	0.12	0.15	0.23	0.41	0.76	NA	0.11	0.04	0.10	0.28	0.19	0.05	0.04	0.52	0.04	0.05	0.36	0.68	0.03	0.03	0.20
Median prediction error ¹ (% of total run time)	0.27	0.61	0.73	1.00	1.25	0.51	1.11	1.52	NA	0.41	0.57	0.67	0.43	0.63	0.31	0.80	1.04	0.27	0.25	1.80	0.45	0.50	0.43	1.60
95% percentile prediction error ¹ (min)	0.39	0.51	0.71	0.41	0.51	0.81	1.66	1.79	NA	0.59	0.20	0.34	3.02	0.43	0.17	0.17	3.29	0.09	0.13	0.74	4.33	0.13	0.20	0.89
Max prediction error ² (min)	0.59	0.88	1.12	0.48	0.78	1.11	2.48	2.15	NA	1.89	0.40	0.38	3.73	0.57	0.36	0.28	4.49	0.13	0.15	0.89	5.37	0.63	0.36	1.12
Median width of 95% CI	0.83	0.73	1.34	1.22	1.10	1.91	2.15	3.86	NA	1.25	0.47	0.99	2.94	1.54	0.85	0.48	4.67	0.73	0.93	1.69	8.78	0.33	0.56	1.01
95% percentile of 95% CI width (min)	1.97	1.65	2.42	1.92	1.90	4.48	3.61	6.63	NA	1.89	1.74	1.93	6.83	2.65	1.75	1.73	8.52	1.71	1.71	2.69	15.7	1.14	1.58	1.61
Max width of 95% CI (min)	2.45	1.96	3.33	1.99	2.00	5.43	5.16	7.43	NA	2.10	2.00	2.05	8.92	3.60	1.96	1.97	9.75	2.00	2.00	2.96	17.3	1.90	1.94	1.90
# Flagged compounds ²	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

¹Based on compounds for which the experimental RT is known. NA = non-applicable. ²Compounds are flagged as suspicious by PredRet if the retention time (RT) is considered to be potentially incorrect, when the difference between the experimental RT and predicted RT is $\geq$ twice the distance from the predicted RT to the outer limits of the prediction interval. CI, confidence interval.

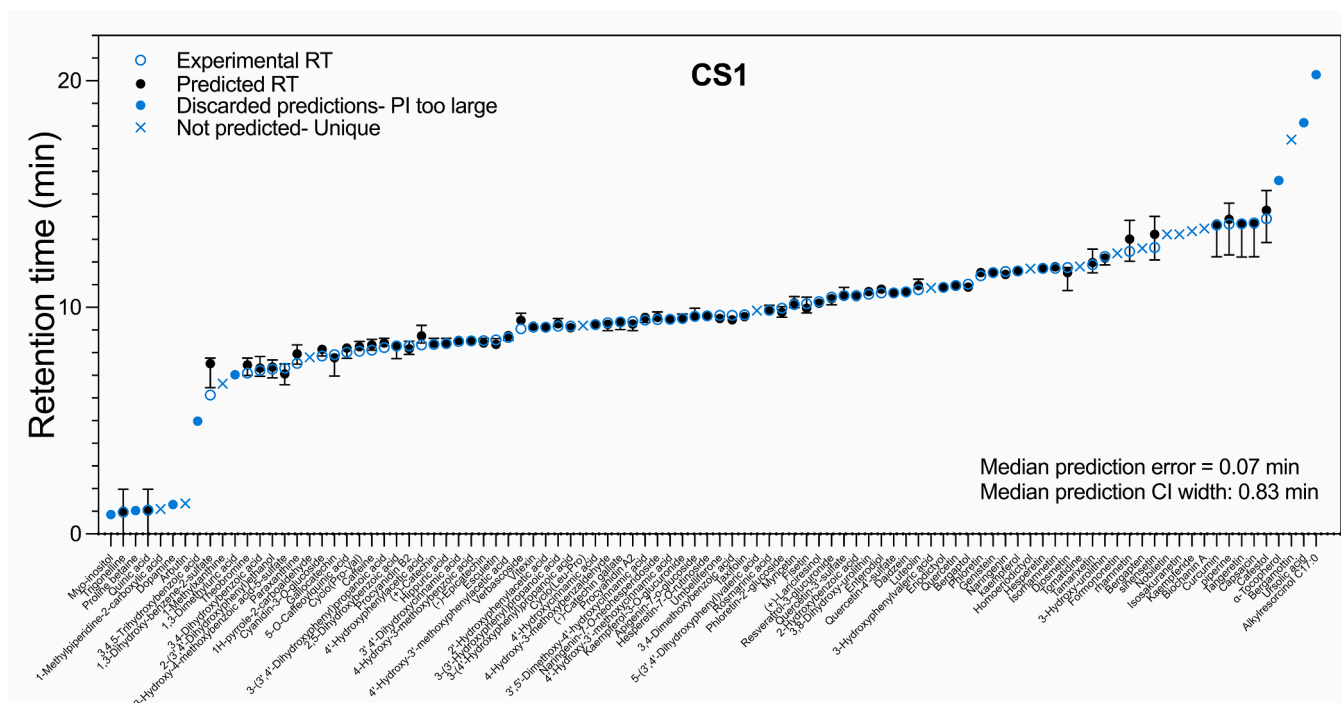


Fig. 3. Retention time (RT) prediction accuracy and coverage of 98 compounds with experimentally known RTs in Chromatographic System (CS) 1. Refer to Supporting Information Tables 3 and 4 for more details of individual compounds.

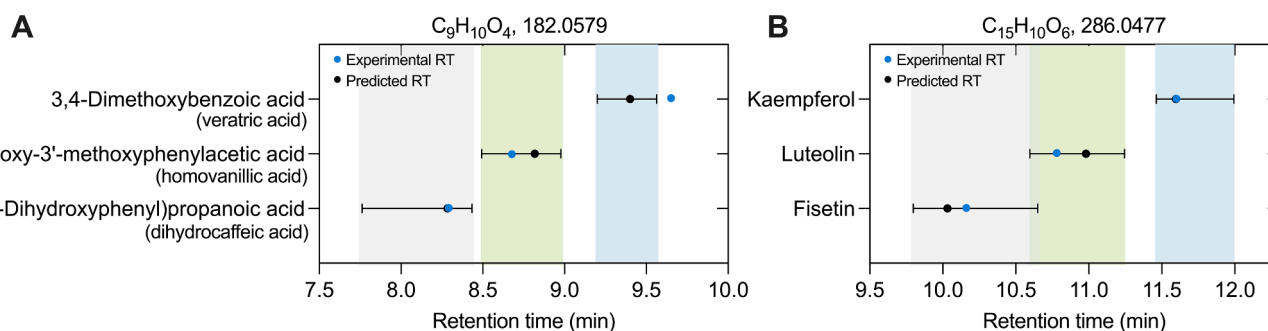


Fig. 4. Retention time (RT) prediction for isomers A) 3,4-dimethoxybenzoic (veratric acid), 4'-hydroxy-3'-methoxyphenylacetic (homovanillic) acid and 3-(3',4'-dihydroxyphenyl)propanoic (dihydrocaffeic) acid, and B) kaempferol, luteolin and fisetin in CS1. Coloured areas represent the prediction interval width.

available. PredRet predictions are precise enough to distinguish structural isomers. Our data sharing initiative and multi-laboratory study contributed to the expansion of the PredRet database with >1500 experimental RTs in 24 CSs for >467 plant food bioactive compounds and their metabolites (>30 families). Importantly, as more experimentally known RTs are entered, more RT predictions are generated and accuracy of the predictions increases. The PredRet database has grown considerably since its introduction and now contains 15,000 RT entries across 68 CSs. Overall, the database covers 4000 unique compounds, beyond plant food bioactive compounds. In comparison, spectral libraries such as the MassBank of North America (MoNA), contain mass spectra for >200,000 compounds, so there remains a large potential for RT sharing. If sufficiently developed to allow accurate RT prediction in any CSs, PredRet would facilitate comparisons between-studies and minimise the need to develop a consensus LC-MS method for plant food compounds. We thus invite the scientific community to contribute to the community-driven open access PredRet database as part of the global effort for annotation of the dark matter of metabolomes. We suggest that sharing of RT as well as collisional cross section data should be as commonplace in the future as sharing of MS/MS data to provide enough

orthogonal data for unambiguous identification in metabolomics.

CRediT authorship contribution statement

Dorrian Yanwen Low: Conceptualization, Formal analysis, Investigation, Visualization, Data curation, Writing - original draft, Supervision. **Pierre Micheau:** Software, Formal analysis, Investigation, Visualization, Data curation, Writing - review & editing. **Ville Mikael Koistinen:** Resources, Writing - review & editing. **Kati Hanhineva:** Resources, Writing - review & editing. **László Abrankó:** Resources, Writing - review & editing. **Ana Rodriguez-Mateos:** Resources, Writing - review & editing. **Andreia Bento da Silva:** Resources, Writing - review & editing. **Christof van Poucke:** Resources, Writing - review & editing. **Conceição Almeida:** Resources, Writing - review & editing. **Cristina Andres-Lacueva:** Resources, Writing - review & editing. **Dilip K. Rai:** Resources, Writing - review & editing. **Esra Capanoglu:** Resources, Writing - review & editing. **Francisco A. Tomás Barberán:** Resources, Writing - review & editing. **Fulvio Mattivi:** Resources, Writing - review & editing. **Gesine Schmidt:** Resources, Writing - review & editing. **Gözde Gürdeniz:** Resources, Writing - review & editing. **Katerina**

Valentová: Resources, Writing - review & editing. **Letizia Bresciani:** Resources, Writing - review & editing. **Lucie Petrášková:** Resources, Writing - review & editing. **Lars Ove Dragsted:** Resources, Writing - review & editing. **Mark Philo:** Resources, Writing - review & editing. **Marynka Ulaszewska:** Resources, Writing - review & editing. **Pedro Mena:** Resources, Writing - review & editing. **Raúl González-Domínguez:** Resources, Writing - review & editing. **Rocío García-Villalba:** Resources, Writing - review & editing. **Senem Kamiloglu:** Resources, Writing - review & editing. **Sonia de Pascual-Teresa:** Resources, Writing - review & editing. **Stéphanie Durand:** Resources, Writing - review & editing. **Wiesław Wiczowski:** Resources, Writing - review & editing. **Maria Rosário Bronze:** Resources, Writing - review & editing. **Jan Stanstrup:** Conceptualization, Software, Formal analysis, Investigation, Data curation, Writing - review & editing. **Claudine Manach:** Conceptualization, Formal analysis, Investigation, Visualization, Data curation, Writing - original draft, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2021.129757>.

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